Homework 0

Due : Sunday, March 1 at 11 : 59 pm

Honor Code

Write and sign the following statement:

> "I certify that all solutions in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted."

Linear Algebra

1. **System of equations (3 points).** Consider the linear system

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 8 \\ 4 & 8 & 12 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 18 \end{bmatrix}.$$

How many solutions does this system have? Explain your reasoning.

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 1 & 8 & 12 \\ 4 & 8 & 12 & 18 \end{array}\right] \xrightarrow{R_2 = R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -3 & 2 & 0 \\ 4 & 8 & 12 & 18 \end{array}\right]$$

$$\xrightarrow{R_3 = R_3 - 4R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -3 & 2 & 0 \\ 0 & 0 & 0 & -6 \end{array}\right]$$

There is no solution because

$0 \neq -6$

2. **Asymptotic powers of 2×2 matrices (6 points).** For each matrix below, determine the behavior of

$$\lim_{n \to \infty} M^n$$

Hint: Use the eigen-decomposition $M = PDP^{-1}$, where $D = \{\lambda_1, \lambda_2, ...\}$ is a diagonal matrix of the eigenvalues. What would be the formula for M^n?

(a) $M_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

(b) $M_2 = \begin{bmatrix} 2 & -5 \\ 1/2 & -7/6 \end{bmatrix}$.

(c) $M_3 = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$.

a) $\begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix}$ $\lambda^2 - 1 = 0$

$\lambda_1 = 1 \quad \lambda_2 = -1$

$\lambda_1 = 1 \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad -x_1 + x_2 = 0 \quad x_1 = x_2$

$\lambda_2 = -1 \quad \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad x_1 + x_2 = 0 \quad x_1 = -x_2$

$M_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = P D P^{-1}$

$D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \rightarrow \lim_{n \to \infty} \begin{bmatrix} 1^n & 0 \\ 0 & (-1)^n \end{bmatrix}$

→ for $n = 2k \quad k \in \mathbb{Z} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

→ for $n = 2k+1 \quad k \in \mathbb{Z} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

there is no limit Diverges

b) $M_2 = \begin{bmatrix} 2 & -5 \\ \frac{1}{2} & -\frac{7}{6} \end{bmatrix} \rightarrow \begin{bmatrix} 2-\lambda & -5 \\ \frac{1}{2} & -\frac{7}{6}-\lambda \end{bmatrix}$

$(\lambda-2)(\lambda+\frac{7}{6}) + \frac{5}{2} = \lambda^2 - \frac{5}{6}\lambda + \frac{1}{6} = 0$

$6\lambda^2 - 5\lambda + 1 = 0 \qquad (3\lambda-1)(2\lambda-1) = 0$

$\lambda_1 = \frac{1}{3} \qquad \lambda_2 = \frac{1}{2}$

$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \qquad \begin{bmatrix} 10 \\ 3 \end{bmatrix}$

$\begin{bmatrix} \frac{5}{3} & -5 \\ \frac{1}{2} & -\frac{9}{6} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$\frac{5}{3}x_1 - 5x_2 = 0$

$x_1 = 3x_2 \qquad x_1 = 3k \qquad x_2 = k$

$\begin{bmatrix} \frac{3}{2} & -5 \\ \frac{1}{2} & -\frac{10}{6} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$\frac{3}{2}x_1 - 5x_2 = 0$

$\frac{3}{2}x_1 = 5x_2 \qquad x_1 = 10k \qquad x_2 = 3k$

$P = \begin{bmatrix} 3 & 10 \\ 1 & 3 \end{bmatrix} \qquad P^{-1} = \frac{1}{9-10}\begin{bmatrix} 3 & -10 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} -3 & 10 \\ 1 & -3 \end{bmatrix}$

$M_2 = \begin{bmatrix} 3 & 10 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -3 & 10 \\ 1 & -3 \end{bmatrix}$

$\lim_{n\to\infty} \begin{bmatrix} 3 & 10 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{3}^n & 0 \\ 0 & \frac{1}{2}^n \end{bmatrix} \begin{bmatrix} -3 & 10 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ Converges

c) $A_3 = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$ $\begin{bmatrix} 4-\lambda & -2 \\ 1 & 1-\lambda \end{bmatrix}$

$(\lambda-1)(\lambda-4)+2 = \lambda^2-5\lambda+6 = (\lambda-3)(\lambda-2)$

$\lambda_1 = 3$ $\lambda_2 = 2$

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$x_1 = 2x_2$ $x_1 = 2t$ $x_2 = t$

$\begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$ $x_1 = x_2$

$P = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ $P^{-1} = \frac{1}{1}\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$

$A_3 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$

$\lim_{n\to\infty} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 3^n & 0 \\ 0 & 2^n \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \infty$ (Diverges)

3. **Singular Value Decomposition (4 points).** Given the matrix

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$$

Find a unit vector x ($\|x\| = 1$) for which $\|Ax\|$ is maximized.

Hint: Recall from SVD that $A = U\Sigma V^\top$, where A maps each right singular vector (a column of V) to a left singular vector (a column of U), scaled by the corresponding singular value. For which right singular vector does this scaling make $\|Ax\|$ the largest?

$$A^TA = \begin{bmatrix} 2 & 2 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix} \quad \begin{bmatrix} 8-\lambda & 2 \\ 2 & 5-\lambda \end{bmatrix}$$

$$(\lambda-8)(\lambda-5)-4 = \lambda^2-13\lambda+36 = (\lambda-9)(\lambda-4)=0$$

$\boxed{\lambda_1 = 9} > \lambda_2 = 4$

$\sigma_1 = 3 \quad \sigma_2 = 2$

$$\begin{bmatrix} 8-9 & 2 \\ 2 & 5-9 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}\frac{1}{\sqrt{5}}$$

$$-x_1+2x_2 = 0$$

$$x_1 = 2x_2$$

so $\|Ax\|$ can be maximum 3

for $x = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$

Calculus

4. **Partial Derivatives (8 points). (First and second derivatives)**

   (a) For $f(x_1, x_2) = x_1^3 + x_2^3 - 3x_1x_2$, find all the first and second order partial derivatives.

   (b) Let $f(x, y) = 4x^2 + y^2 - 8xy + 4x + 6y - 10$.
   Find the critical points by solving

   $$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

   simultaneously and determine which point(s) yield a minimum value.

   (c) For $f(x, y) = e^{xy} + x^2y$, compute:
   (i) $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$
   (ii) $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$, and $\frac{\partial^2 f}{\partial x \partial y}$

   (d) For $f(x, y) = \ln(x^2 + y^2 + 1)$, compute:
   (i) $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$
   (ii) $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$, and $\frac{\partial^2 f}{\partial x \partial y}$

a) $\frac{\partial f(x_1, x_2)}{\partial x_1} = 3x_1^2 - 3x_2$

$\frac{\partial^2 f(x_1, x_2)}{\partial x_1^2} = 6x_1$

$\frac{\partial f(x_1, x_2)}{\partial x_2} = 3x_2^2 - 3x_1$

$\frac{\partial^2 f(x_1, x_2)}{\partial x_2^2} = 6x_2$

$\frac{\partial^2 f(x_1, x_2)}{\partial x_1 \partial x_2} = -3$

b) $\frac{\partial f}{\partial x} = 8x - 8y + 4$ $\quad \frac{\partial f}{\partial y} = 2y - 8x + 6$

$8x - 8y + 4 = 0$ $\quad y = \frac{5}{3}$ $\quad x = \frac{7}{6}$

$+ \; 2y - 8x + 6 = 0$ $\quad \left(\frac{7}{6}, \frac{5}{3}\right)$

$-6y + 10 = 0$

$D = f_{xx} f_{yy} - (f_{xy})^2$

$f_{xx} = 8 \quad f_{yy} = 2 \quad f_{xy} = -8$

$8 \cdot 2 - (8)^2 = -48$

$D < 0$ so $\left(\frac{7}{6}, \frac{5}{3}\right)$ is saddle point there is minimum value for this function

c) $\frac{\partial f}{\partial x} = y \cdot e^{xy} + 2xy$ $\quad \frac{\partial f}{\partial y} = x \cdot e^{xy} + x^2$

$\frac{\partial^2 f}{\partial x^2} = y^2 \cdot e^{xy} + 2y$ $\quad \frac{\partial^2 f}{\partial y^2} = x^2 e^{xy}$

$\frac{\partial^2 f}{\partial x \partial y} = e^{xy} + xy e^{xy} + 2x$

d) $\frac{\partial f}{\partial x} = \frac{1}{x^2+y^2+1} \cdot 2x$ $\quad \frac{\partial f}{\partial y} = \frac{1}{x^2+y^2+1} \cdot 2y$

$$\frac{\partial^2 f}{\partial x^2} = \frac{2(x^2+y^2+1) - 2x\,2x}{(x^2+y^2+1)^2} = \frac{-2x^2+2y^2+2}{(x^2+y^2+1)^2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{2(x^2+y^2+1) - 2y\,2y}{(x^2+y^2+1)^2} = \frac{2x^2-2y^2+2}{(x^2+y^2+1)^2}$$

$$\frac{\partial^2 f}{\partial x \partial y} = -\frac{1}{(x^2+y^2+1)^2} \cdot 2y \cdot 2x = \frac{-4xy}{(x^2+y^2+1)^2}$$

5. **Recursive expression and derivatives (6 points).** Suppose we have variables $z_1, z_2, \ldots, z_n$, $w_1, \ldots, w_{n-1}$, and $b_1, \ldots, b_{n-1} \in \mathbb{R}$, where

$$z_n = w_{n-1} z_{n-1} + b_{n-1}.$$

(a) Compute $\frac{dz_k}{dz_{k-1}}$ for $k = 2, \ldots, n$.

(b) Compute $\frac{dz_n}{dz_1}$.

(c) Compute $\frac{dz_n}{db_1}$. You may express your answers in terms of $z_1, ..., z_n, w_1, ..., w_{n-1}, b_1, ..., b_{n-1}$

a) $\frac{\partial z_k}{\partial z_{k-1}} = w_{k-1}$

$k = 2 \quad w_1$

$k = 3 \quad w_2$

$\vdots$

$k = n \quad w_{n-1}$

b) $\frac{\partial z_n}{\partial z_1} = \frac{\partial z_n}{\partial z_{n-1}} \cdot \frac{\partial z_{n-1}}{\partial z_{n-2}} \cdots \frac{\partial z_2}{\partial z_1}$

$= w_{n-1} \cdot w_{n-2} \cdots w_1$

c) $\frac{\partial z_n}{\partial b_{n-1}} = 1 \qquad \frac{\partial z_2}{\partial b_1} = (w_1 z_1 + b_1)' = 1$

$\frac{\partial z_n}{\partial b_1} = \frac{\partial z_n}{\partial z_{n-1}} \cdot \frac{\partial z_{n-1}}{\partial z_{n-2}} \cdots \frac{\partial z_3}{\partial z_2} \cdot \frac{\partial z_2}{\partial b_1}$

$= w_{n-1} \cdot w_{n-2} \cdots w_2 \cdot 1$

Probability

6. **Conditioned uniform difference (3 points).** Let $X, Y \overset{\text{iid}}{\sim} \text{Unif}(-1, 1)$. Compute

$$\mathbb{P}\big(|X - Y| \leq 0.5 \,\big|\, XY > 0\big).$$

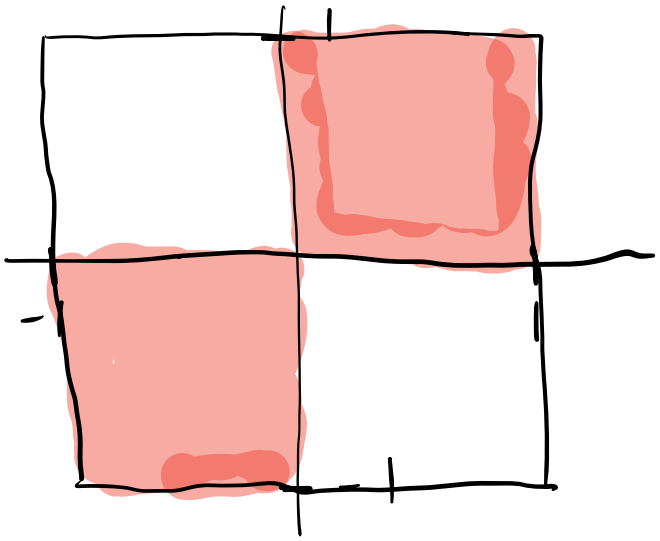
Hint: Try drawing a 2D cartesian plane where the horizontal and vertical axes represent X and Y respectively and each range from -1 to 1. For which quadrants is it true that $XY > 0$? Within these quadrants, how can we visualize the region $|X - Y| < 0.5$?

$A = |X - Y| \leq 0.5 \qquad B = XY > 0$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

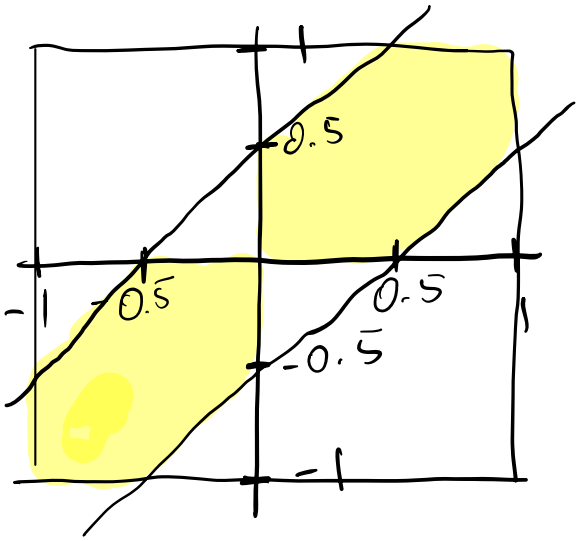
$XY > 0$

$|X - Y| \leq 0.5$



$P(B) = 2$

$P(A \cap B) = 2 \cdot (1 - 0.25)$

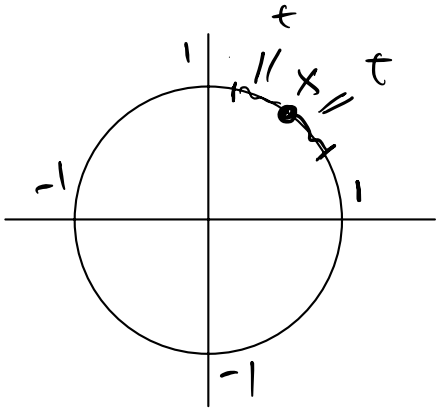


$$P(A|B) = \frac{1.5}{2} = \frac{3}{4}$$

7. **Nearest–neighbor arc length (4 points).** Suppose you select 20 i.i.d. points $X_1, \ldots, X_{20}$ uniformly at random on the circumference of the unit circle.

   (a) Let D be the shortest arc distance from X_1 to the nearest of the other 19 points. Calculate $\mathbb{P}(D > t)$ where $0 \le t \le \frac{1}{2}$.
   Hint: What does the event $\{D > t\}$ mean in terms of where the other 19 points can be located relative to X_1?

   (b) Find the expected arc length (in degrees) between X_1 and the point nearest to it.
   Hint: The Wikipedia page on expected value of continuous variables *may be helpful here.*



a) Let x be the arc distance from X_1 to other point

$$P(x > t) = \frac{2\pi - 2t}{2\pi}$$

so

$$P(D > t) = \left(\frac{2\pi - 2t}{2\pi}\right)^{19}$$

b) $$E[D] = \int_0^{\pi} \left(1 - \frac{t}{\pi}\right)^{19} dt \qquad 1 - \frac{t}{\pi} = v$$

$$-\frac{1}{\pi} dt = dv$$

$$-\pi \int_1^0 v^{19} dv = -\pi \frac{v^{20}}{20}\Big|_{v=1}^{v=0} = \frac{\pi}{20}$$

$$E[D] = \frac{\pi}{20}$$

8. **Cancer screening (3 points).** A medical test has sensitivity 90% and false–positive rate 3%:

$$\mathbb{P}(T = 1 \mid C = 1) = 0.9, \qquad \mathbb{P}(T = 1 \mid C = 0) = 0.03.$$

Suppose the disease prevalence is very low, $\mathbb{P}(C = 1) = 0.001$. Compute the posterior probability of disease given a positive test:

$$\mathbb{P}(C = 1 \mid T = 1).$$

$$P(T=1 \mid C=1) = \frac{P(T=1 \cap C=1)}{P(C=1)} = 0.9$$

$$P(C=1) = 0.001 \quad \text{so} \quad P(T=1 \cap C=1) = 0.0009$$

$$P(T=1 \mid C=0) = \frac{P(T=1 \cap C=0)}{P(C=0)} = 0.03$$

$$P(C=0) = 0.999 \quad \text{so} \quad P(T=1 \cap C=0) = 0.02997$$

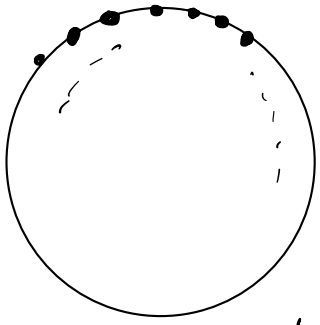
$$P(C=1 \mid T=1) = \frac{P(C=1 \cap T=1)}{P(T=1)}$$

$$P(T=1) = P(T=1 \cap C=1) + P(T=1 \cap C=0)$$

$$P(T=1) = 0.0009 + 0.02997 = 0,03087$$

$$P(C=1 \mid T=1) = \frac{0.00090}{0.03087} = 0,029154519$$

9. **Follower Counts (3 points).** Suppose 420 people are sitting uniformly at random around a circle, each with a *distinct* number of TikTok followers. What is the expected number of people whose follower count is higher than both their immediate neighbors?



X = People whose follower count is higher than both their immediate neighbors

$P(X) = \frac{1}{3}$ because if we choose 3 random number one of them is bigest with $\frac{1}{3}$ probability

$$E[X] = \int_0^{420} \frac{1}{3}\,dx = \frac{x}{3}\Big|_{x=0}^{x=420} = \frac{420}{3} - 0 = 140$$